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⑦ How does identifying a problem as NP-Hard influence the approach or the algorithm development

Answer

An NP-Hard problem is a problem that is at least as hard as the hardest problem in NP.

By this no known polynomial time can solve this problem unless $NP = P$

Identifying a problem as NP Hard changes the approach in that the exact algorithm becomes less practical

Therefore there is to be a shift to approximation or heuristic instead of exact solutions

Common strategies that can be used may include greedy algorithm and local search or randomized algorithms